KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam Practice

Practice Exam 3

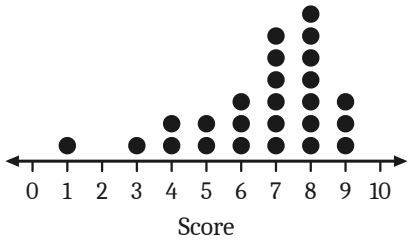
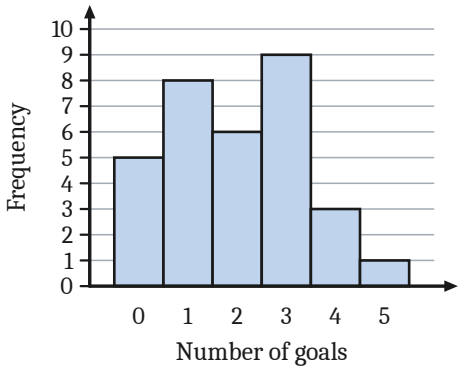
Instructions

1. Working time: 60 minutes
2. Attempt all questions
3. Show ALL working where needed
4. A calculator is permitted
5. Answers are at the back of this booklet

Sections and Marks

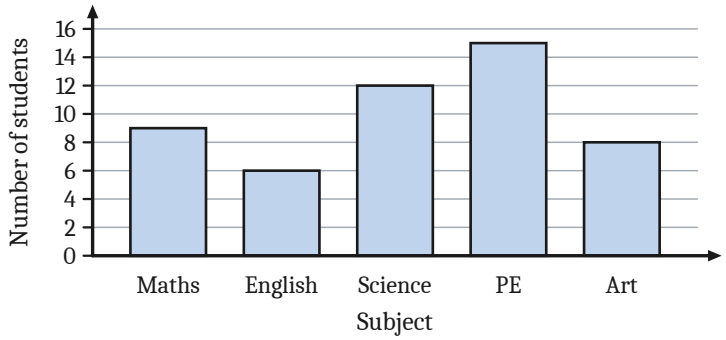
Section 1 - Data	/14
Section 2 - Surface area and volume	/20
Section 3 - Mixed Topics	/30
Total marks	**/64**

Section 1: Data

	Multiple Choice:	Mark
1	Which of the following is **categorical** data? A. The length of a snake B. The number of goals in a soccer game C. Favourite sport D. The temperature at noon	1
2	The dot plot shows the scores (out of 10) in a vocabulary quiz of 25 students. Which statement best describes the data? A. Symmetrical, clustered at 7 and 8 B. Negatively skewed, clustered at 1 and 3 C. Positively skewed, clustered at 7 and 8 D. Negatively skewed, clustered at 7 and 8 	1
3	The histogram shows the number of goals scored by a hockey team in each of 32 games. What is the mode? A. 9 B. 2.5 C. 3 D. 5 	1
4	A family recorded the number of loads of washing they did each week for 9 weeks: 3, 4, 4, 5, 3, 4, 19, 5, 4 Which measure of centre changes the most if the outlier, 19, is removed? A. The median B. The mean C. The mode D. All three change by the same amount	1

Section 1: Data (continued)

Multiple Choice:

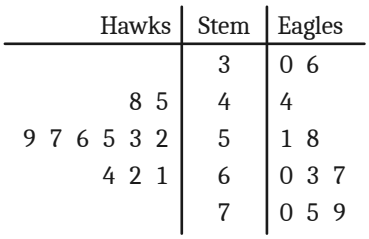
		Mark
5	Year 9 students were asked to name their favourite subject. The column graph shows the results. What percentage of the students chose maths or science?  A. 21% B. 24% C. 58% D. 42%	1

Short Answer:

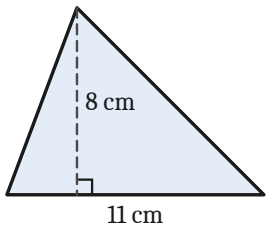
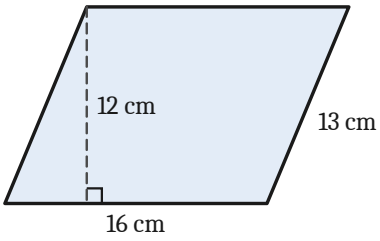
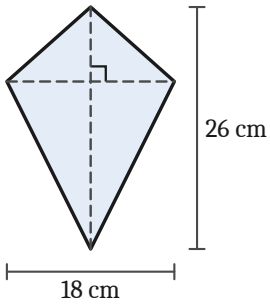
	Question	Working Out	Mark
6	The stem-and-leaf plot shows the heights, in centimetres, of 11 tomato plants. Stem | Leaf 4 | 1 5 8 5 | 2 3 6 8 8 6 | 1 4 9 Key: 5 | 3 = 53 cm a. Find the mean. b. Find the median. Note: Mean = sum of the scores ÷ number of scores	a. b.	2 1

Section 1: Data (continued)

Worded Problems:

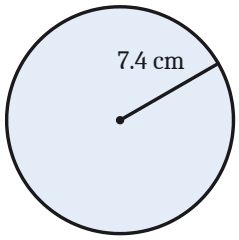
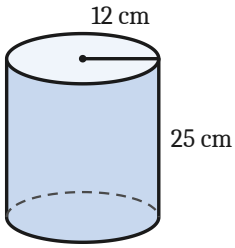
	Question	Working Out	Mark
7	The Hawks and the Eagles are two archery teams. The back-to-back stem-and-leaf plot shows each team's score in 11 rounds. Note: Range = highest score − lowest score	 Key: 5 | 6 = 56 points	
a	Find the median score for each team.		2
b	Which team usually scores higher? Give a reason.		1
c	Which team is more consistent? Find the range of each team to explain.		3
	Total		**/14**

Section 2: Surface area and volume

	Multiple Choice:	Mark
1	A jug holds 4500 mL of juice. How many litres is this? A. 45 L B. 0.45 L C. 4.5 L D. 450 L	1
2	What is the area of this triangle? A. 44 cm^2 B. 88 cm^2 C. 19 cm^2 D. 9.5 cm^2 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 208 cm^2 B. 96 cm^2 C. 58 cm^2 D. 192 cm^2 	1
4	What is the area of this kite? Note: Area of a kite: $A = \frac{1}{2}xy$ (x and y are the diagonals) A. 468 cm^2 B. 117 cm^2 C. 234 cm^2 D. 44 cm^2 	1

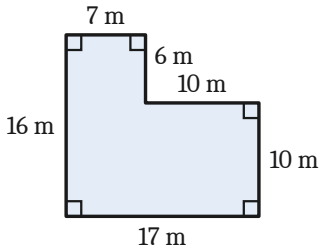
Section 2: Surface area and volume (continued)

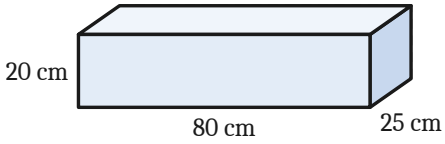
Short Answer:

	Question	Working Out	Mark
5	For this circle, find, correct to 1 decimal place:  a. its circumference b. its area. Note: Circumference: $C = 2\pi r$ Note: Area: $A = \pi r^2$	a. b.	1 1
6	A closed rubbish bin is a cylinder. It has a radius of 12 cm and a height of 25 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. b.	2 2

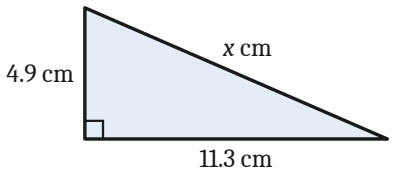
Section 2: Surface area and volume (continued)

Worded Problems:

	Question	Working Out	Mark
7	The plan shows the floor of an L-shaped hall. All of the corners are right angles. Note: Area of a rectangle: $A = lb$ 		
a	Skirting board is fitted around the edge of the floor. How many metres of skirting board are needed?		1
b	Find the area of the floor.		3
c	Carpet costs $38.90 per square metre. Find the cost of carpeting the whole floor.		1

	Section 2: Surface area and volume (continued)		
	Question	**Working Out**	**Mark**
8	A closed planter box is shaped like a box. It is a rectangular prism 80 cm long, 25 cm wide and 20 cm high. 		
a	Find the volume of the box in cm^3. Note: Volume of a rectangular prism: $V = lwh$		1
b	How many litres does the box hold? Note: 1 L = 1000 cm^3		1
c	The outside of the planter box is to be stained. Find the surface area of the box. Note: Surface area = the total area of the 6 faces; area of a rectangle: $A = lb$		3
	Total		**/20**

Section 3: Mixed Topics (Chapters 1–8)

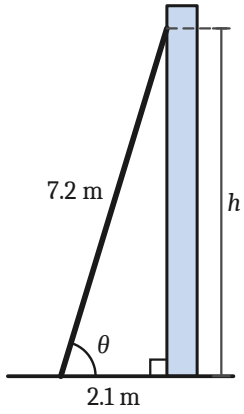
	Multiple Choice:	Mark
1	If $m = 3$ and $n = 5$, evaluate $4m^2 - n$. A. 7 B. 19 C. 31 D. 139	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 16.2 B. 12.3 C. 10.2 D. 6.4 	1
3	A scooter is marked at \$150. In a sale it is discounted by 30%. What is the selling price? A. \$195.00 B. \$120.00 C. \$105.00 D. \$45.00	1
4	Simplify $2a^6 \times 7a^3$. A. $14a^3$ B. $14a^9$ C. $9a^9$ D. $14a^{18}$	1
5	Ana, an engineer, earns an annual salary of \$91 260. How much is this per fortnight? Note: 1 year = 52 weeks = 26 fortnights = 12 months A. \$1755.00 B. \$3510.00 C. \$3802.50 D. \$7605.00	1

Short Answer:

	Question	Working Out	Mark
6	Simplify: a. $3x + 8y - x - 3y$ b. $21pq \div 7p$	a. b.	1 1
7	Expand and simplify: $2(5a - 1) - 4(a - 3)$		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
8	Factorise: a. $10y + 25$ b. $x^2 - 4x$	a. b.	1 1
9	Leo invests \$5600 for 3 years at 2.75% p.a. simple interest. Calculate the interest earned. Note: Simple interest: $I = Prn$		2
10	A dealer buys a laptop for \$520 and sells it for \$702. a. Find the profit. b. Find the profit as a percentage of the cost price. Note: Percentage profit = profit ÷ cost price × 100%	a. b.	1 1
11	Solve: $8k - 2(k + 5) = 3k + 8$		3
12	Ruby sells phones. Ruby is paid \$350 per week plus 4% commission on sales. Last week Ruby sold \$27 250 worth of phones. Calculate Ruby's total pay for the week.		2

	Section 3: Mixed Topics (Chapters 1–8) (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
13	Josh works at a cinema. The normal rate of pay is \$25.40 per hour for the first 38 hours each week, and time-and-a-half for each hour after that. Last week Josh worked 42 hours.		
a	Calculate the pay for the first 38 hours.		1
b	Calculate the total pay for the 42 hours.		2
14	Guitar lessons cost a \$50 enrolment fee plus \$35 per lesson. The total cost, C dollars, for n lessons is given by the formula $\boldsymbol{C = 50 + 35n}$		
a	Find the total cost for 8 lessons.		1
b	Ava paid \$610 altogether. Write an equation and solve it to find how many lessons Ava had.		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
15	A ladder 7.2 m long leans against the vertical wall of a house. The foot of the ladder is 2.1 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.		2
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.		2
	Total		**/30**

Answers

Mark your own work. If an answer is wrong, read the help sheet topic in the last column, then try the question again.

Section 1 answers – Data			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (Favourite sport)	1	D1 Types of data
2	D (Negatively skewed, clustered at 7 and 8)	1	D2 Shape of a distribution
3	C (3)	1	D3 Reading graphs
4	B (The mean)	1	D5 Outliers
5	D (42%)	1	D3 Reading graphs
6a	55 cm	2	D4 Mean, median, mode, range
6b	56 cm	1	D4 Mean, median, mode, range
7a	Hawks: 56; Eagles: 60	2	D6 Comparing two data sets
7b	Eagles: the median for Eagles (60) is higher than the median for Hawks (56).	1	D6 Comparing two data sets
7c	Hawks: the range is 19 (64 – 45), much smaller than the range for Eagles, 49 (79 – 30).	3	D6 Comparing two data sets
	Total	**14**	

Section 2 answers – Surface area and volume			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (4.5 L)	1	M1 Units
2	A (44 cm^2)	1	M2 Area of shapes
3	D (192 cm^2)	1	M2 Area of shapes
4	C (234 cm^2)	1	M2 Area of shapes
5a	46.5 cm	1	M3 Circles
5b	172.0 cm^2	1	M3 Circles
6a	11 309.7 cm^3	2	M5 Volume
6b	2789.7 cm^2	2	M6 Surface area
7a	66 m	1	M4 Composite shapes
7b	212 m^2	3	M4 Composite shapes
7c	$8246.80	1	M4 Composite shapes
8a	40 000 cm^3	1	M5 Volume
8b	40 L	1	M1 Units
8c	8200 cm^2	3	M6 Surface area
	Total	**20**	

Section 3 answers – Mixed Topics (Chapters 1–8)			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (31)	1	A1 Substitution
2	B (12.3)	1	T1 Pythagoras
3	C ($105.00)	1	F1 Percentages and profit
4	B ($14a^9$)	1	A5 Index laws
5	B ($3510.00)	1	F3 Earning money
6a	$2x + 5y$	1	A2 Simplifying
6b	$3q$	1	A2 Simplifying
7	$6a + 10$	2	A3 Expanding
8a	$5(2y + 5)$	1	A4 Factorising
8b	$x(x - 4)$	1	A4 Factorising
9	$462.00	2	F2 Simple interest
10a	$182	1	F1 Percentages and profit
10b	35%	1	F1 Percentages and profit
11	$k = 6$	3	E1 Solving equations
12	$1440	2	F3 Earning money
13a	$965.20	1	F3 Earning money
13b	$1117.60	2	F3 Earning money
14a	$330	1	E2 Formulas and word problems
14b	16 lessons	2	E2 Formulas and word problems
15a	6.89 m	2	T1 Pythagoras
15b	73°	2	T2 Trigonometry
	Total	**30**	